Some Computational QM Principles - Project 2

CURIE SP26 - Programming in Computational Quantum Chemistry

The first two sections will be based on an existing code from jupyter notebook cells. (You may run these as a single python script as well) Significant Portions of this project was inherited from class material from CHEM 121.

1 Variational Principle

1.1 1D Harmonic Oscillator (Background)

The 1D Quantum Harmonic Oscillator:

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+\frac{1}{2}m\omega^2x^2\right)\psi(x)=E\psi(x)$$

has the energy and wavefunctions:

$$E=\hbar\omega\left(n+\frac{1}{2}\right) \qquad \psi_n(x)=\frac{1}{\sqrt{2^n n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}e^{-\frac{m\omega x^2}{2\hbar}}H_n\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right)$$

Where $H_n(x)$ is the Hermite polynomial. To simplify the problem we remove all constants and units by defining $x^*=x/l$ and $\hat{H}^*=\hat{H}/\epsilon$ where $\epsilon=\hbar^2/ml^2=m\omega^2l^2$. The reduced Hamiltonian and its solutions are:

$$\hat{H}^*=-\frac{1}{2}\frac{d^2}{dx^2}+\frac{1}{2}(x^*)^2$$

$$E_n^*=n+\frac{1}{2} \qquad \psi_n(x^*)=\frac{1}{\sqrt{2^n n!}}\pi^{-1/4}e^{-\frac{(x^*)^2}{2}}H_n(x^*)$$

with the first three energies and wavefunctions shown in Table 1.1.

n	E_n^*	$\psi_n(x^*)$
0	$\frac{1}{2}$	$\pi^{-1/4} e^{-\frac{(x^*)^2}{2}}$
1	$\frac{3}{2}$	$\pi^{-1/4}\sqrt{2}\, x^*\, e^{-\frac{(x^*)^2}{2}}$
2	$\frac{5}{2}$	$\pi^{-1/4}\frac{1}{\sqrt{2}}\left(2(x^*)^2 - 1\right) e^{-\frac{(x^*)^2}{2}}$

Table 1: First three energy and wavefunctions of the harmonic oscillator in reduced units.

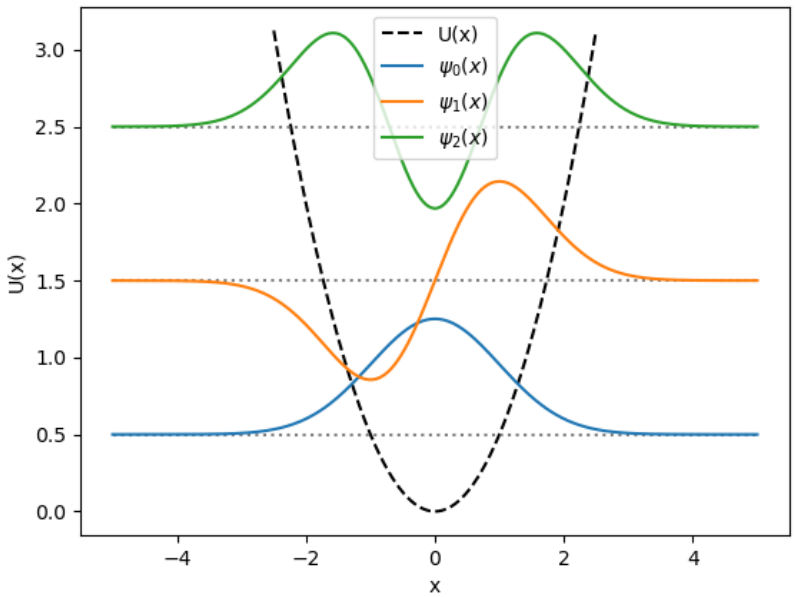

Figure 1: Plot of the first three harmonic oscillator wavefunctions.

1.2 Approximating with the Linear Variational Principle

I recommend reading Section 4.1 on the variational principle. We will approximate the wavefunction and solution of the first three wavefunctions of the Harmonic oscillators with a linear combination of gaussian functions serving as our basis of "orbitals":

$$\psi(x) = \sum_{\mu}^{n} c_\mu \phi_\mu(x) \quad \text{where} \quad \phi_\mu(x) = \exp[-\alpha(x - x_a)^2] \qquad \alpha = 2$$

The gaussian functions are evenly spaced and offset from the center x by x_A with a fixed width parameter $\alpha = 2$. We will use a different number of basis functions ($n = 5, 10, 15, 20$) and compare their accuracy in approximating the energy and wavefunctions.

For computing the matrix elements/integrals, the analytical solutions are given as:

$$
\begin{aligned}
S_{\mu\nu} = \langle \phi_\mu | \phi_\nu \rangle &= \int_{-\infty}^{+\infty} dx \exp\left(-\alpha(x - x_A)^2\right) \exp\left(-\alpha(x - x_B)^2\right) \\
&= \sqrt{\frac{\pi}{2\alpha}} \exp\left(-\frac{\alpha}{2}(x_A - x_B)^2\right) \\
H_{\mu\nu} = \langle \phi_\mu | H | \phi_\nu \rangle &= \int_{-\infty}^{+\infty} dx \exp\left(-\alpha(x - x_A)^2\right) \left(-\frac{d^2}{dx^2} + \frac{1}{2}x^2\right) \exp\left(-\alpha(x - x_B)^2\right) \\
&= \frac{1}{2} S_{\mu\nu} \left[\alpha - \alpha^2 (x_A - x_B)^2 + \frac{1}{4}(\frac{1}{2} + (x_A + x_B)^2)\right] \\
&= \frac{1}{2} \left[\sqrt{\frac{\pi}{2\alpha}} \exp\left(-\frac{\alpha}{2}(x_A - x_B)^2\right)\right] \left[\alpha - \alpha^2 (x_A - x_B)^2 + \frac{1}{4}(\frac{1}{2} + (x_A + x_B)^2)\right]
\end{aligned}
$$

with the overlap and the Hamiltonian matrices formulated as:

$$
\mathbf{S} = \begin{bmatrix} \langle \phi_1 | \phi_1 \rangle & \cdots & \langle \phi_1 | \phi_n \rangle \\ \vdots & \ddots & \vdots \\ \langle \phi_n | \phi_1 \rangle & \cdots & \langle \phi_n | \phi_n \rangle \end{bmatrix} \qquad \mathbf{H} = \begin{bmatrix} \langle \phi_1 | \hat{H} | \phi_1 \rangle & \cdots & \langle \phi_1 | \hat{H} | \phi_n \rangle \\ \vdots & \ddots & \vdots \\ \langle \phi_n | \hat{H} | \phi_1 \rangle & \cdots & \langle \phi_n | \hat{H} | \phi_n \rangle \end{bmatrix}
$$

Here are some more explicit instructions:

- Implement code that evaluates $S_{\mu\nu}$ and $H_{\mu\nu}$ given the analytical solution of the integrals above and build $\mathbf{S}$ and $\mathbf{H}$ with `n_basis = [5, 10, 15, 20]` gaussian basis functions evenly by x_A in the range $[-5, 5]$ and a fixed width parameters $\alpha = 2$.
- Build the orthogonalizing transformation $\mathbf{X}$ with the $\mathbf{S}$ you built (read Section 4.2) and orthogonalize your Hamiltonian $\mathbf{H}' = \mathbf{X}^\mathbf{T}\mathbf{H}\mathbf{X}$.
- Evaluate the energy and variationally optimized coefficients by diagonalizing $\mathbf{H}'$ (remember the transform the coefficients back into the original basis $\mathbf{c} = \mathbf{X}\mathbf{c}'$). Compare the accuracy of the energy of the first three states.
- Build the first three approximate HO wavefunctions by taking linear combinations of the variationally optimized coefficients and its corresponding gaussian basis functions and plot it over the original exact wavefunctions. Repeat for each `n_basis` and compare their accuracy. (The resulting function may be flipped randomly, just plot the inverse of the function in that case)

2 Mean Field Theory and Correlation

We will now consider a system where two wavefunctions are coupled by considering the coupled 2D Harmonic Oscillator defined by the Hamiltonian:

$$\hat{H}^* = -\frac{d^2}{dx^2} + \frac{1}{2}x^2 - \frac{d^2}{dy^2} + \frac{1}{2}y^2 + ax^4y^4 = h(x^*) + h(y^*) + ax^4y^4$$

2.1 2D Harmonic Oscillator (Background)

Before the coupled oscillator system, consider a uncoupled 2D HO system where $\hat{H} = h(x) + h(y)$. To solve this problem, we deployed the technique separation of variables $\Psi(x, y) = \psi(x)\psi(y)$, where the particle in x and y are treated independently:

$$\begin{aligned} E &= \int dx \int dy\ \Psi^*(x,y)\hat{H}\Psi(x,y) \\ &= \int dx \int dy\ \psi^*(x)\psi^*(y)\hat{H}\psi(x)\psi(y) \qquad \text{(integral form)} \end{aligned}$$

$$\begin{aligned} E &= \langle\Psi(x,y)|\hat{H}|\Psi(x,y)\rangle \qquad\qquad \text{(bra-ket notation)} \\ &= \langle\psi(x)|\langle\psi(y)|\hat{H}|\psi(x)\rangle|\psi(y)\rangle \\ &= \langle\psi(x)|\langle\psi(y)|\left[h(x) + h(y)\right]|\psi(x)\rangle|\psi(y)\rangle \\ &= \langle\psi(x)|h(x)|\psi(x)\rangle\langle\psi(y)|\psi(y)\rangle + \langle\psi(x)|\psi(x)\rangle\langle\psi(y)|h(y)|\psi(y)\rangle \\ &= \langle\psi(x)|h(x)|\psi(x)\rangle + \langle\psi(y)|h(y)|\psi(y)\rangle \\ &= E^*_{n_x} + E^*_{n_y} \end{aligned}$$

Note that $\langle\psi(y)|\psi(y)\rangle = \langle\psi(x)|\psi(x)\rangle = 1$ due to orthogonality. We can show that the particles are indeed independent for this Ansatz by computing the probability density at any point $x, y \in \mathbb{R}$:

$$\begin{aligned} |\Psi(x,y)|^2 &= \int dx \int dy\ \langle\Psi(x,y)|\Psi(x,y)\rangle \\ &= \int dx \int dy\ \langle\psi(x)|\langle\psi(y)|\psi(x)\rangle|\psi(y)\rangle \\ &= \int dx\ \langle\psi(x)|\psi(x)\rangle \int dy\ \langle\psi(y)|\psi(y)\rangle \\ &= |\psi(x)|^2\ |\psi(y)|^2 \end{aligned}$$

This works well in cases like this where x and y are separable in the Hamiltonian, you can see that the product wavefunction acts only on the respective Hamiltonian, resulting in separable eigenenergies.

2.2 Mean Field Approximation

Due to the coupling term ax^4y^4, the motion of the in particle $\psi(x)$ now influences the particle $\psi(y)$ and vice. Therefore the energies cannot be variationally minimized independently unlike the normal 2D HO. The simplest approach to minimizing the energy is to use the factorized wavefunction above as our Ansatz [commonly referred to as the Hartree Product: $\Psi_{HP} = \psi(x)\psi(y)$] and averaging out/evaluating the expectation value of the other coupling wavefunction as an external field, known as the mean-field approximation. The rough idea and the corresponding formulas will be outlined below.

First, the Hartree Product wavefunction the energy of the coupled system evaluates to:

$$\begin{aligned}
E &= \int dx \int dy\ \Psi^*(x,y)\left[\hat{h}(x) + \hat{h}(y) + ax^4y^4\right]\Psi(x,y) \qquad \text{(integral form)} \\
&= \int dx \int dy\ \psi^*(x)\psi^*(y)\left[\hat{h}(x) + \hat{h}(y) + ax^4y^4\right]\psi(x)\psi(y) \\
&\text{(if you expand this further, you will achieve the same solution as below)}
\end{aligned}$$

$$\begin{aligned}
E &= \langle\Psi(x,y)|\left[\hat{h}(x) + \hat{h}(y) + ax^4y^4\right]|\Psi(x,y)\rangle \qquad \text{(bra-ket notation)} \\
&= \langle\psi(x)\psi(y)|\left[\hat{h}(x) + \hat{h}(y) + ax^4y^4\right]|\psi(x)\psi(y)\rangle \\
&= \langle\psi(x)|\hat{h}(x)|\psi(x)\rangle + \langle\psi(y)|\hat{h}(y)|\psi(y)\rangle + \langle\psi(x)\psi(y)|ax^4y^4|\psi(x)\psi(y)\rangle \\
&= E^*_{n_x} + E^*_{n_y} + a\langle\psi(x)|x^4|\psi(x)\rangle\langle\psi(y)|y^4|\psi(y)\rangle
\end{aligned}$$

Next, to further seperate the coupling term, the expectation value of other particle is evaluated out:

$$\begin{aligned}
\epsilon_x &= \langle\psi(x)|\left[\hat{h}(x) + ax^4\langle\psi(y)|y^4|\psi(y)\rangle\right]|\psi(x)\rangle \\
&= \langle\psi(x)|\left[\hat{h}(x) + ax^4\langle y^4\rangle\right]|\psi(x)\rangle \\
\epsilon_y &= \langle\psi(y)|\left[\hat{h}(x) + a\langle\psi(x)|x^4|\psi(x)\rangle y^4\right]|\psi(y)\rangle \\
&= \langle\psi(y)|\left[\hat{h}(x) + a\langle x^4\rangle y^4\right]|\psi(y)\rangle
\end{aligned}$$

Here, the operator $ax^4\langle y^4\rangle$ can be interpreted as the operator x^4 with the influence from the probability distribution or the average "field" of particle y, $\langle y^4\rangle$, and vice versa for $a\langle x^4\rangle y^4$, hence the name mean-field approximation. Defining the effective Hamiltonian $h_{eff}(x)$ and

$h_{eff}(y)$ as the operators above, we can transform this problem into a system of coupled eigenvalue problems:

$$\hat{h}_{eff}(x) = h(x) + ax^4\langle y^4\rangle$$
$$\hat{h}_{eff}(y) = h(y) + a\langle x^4\rangle y^4$$

$$\hat{h}_{eff}(x)\psi(x) = \epsilon_1\psi(x)$$
$$\hat{h}_{eff}(y)\psi(y) = \epsilon_2\psi(y)$$

The mathematical manipulation here was ad hoc for explaing the mean-field approximation. The proper contained minimization procedure is outlined in Section 4.4.

Now expanding these equation into our linear combination of gaussian function basis, the matrix elements/integral for the coupling operator x^4 evaluates to:

$$G_{\mu\nu} = \langle\phi_\mu|x^4|\phi_\nu\rangle = S_{\mu\nu}\left[\frac{3}{16\alpha^2} + \frac{3}{8\alpha}(x_A + x_B)^2 + \frac{1}{16}(x_A + x_B)^4\right]$$

with the resulting coupled eigenvalue system below. For the total energy, ϵ_1 and ϵ_2 each carries a $a\langle x^4\rangle\langle y^4\rangle$ which results in double counting. To address this, a factor of $a\langle x^4\rangle\langle y^4\rangle$ is removed.

$$\left(\mathbf{H} + a(\mathbf{C}^\dagger{}_y\mathbf{G}\mathbf{C}_y)\mathbf{G}\right)\mathbf{C}_x = \epsilon_1\mathbf{S}\mathbf{C}_x \qquad \langle y^4\rangle = \mathbf{C}^\dagger{}_y\mathbf{G}\mathbf{C}_y$$
$$\left(\mathbf{H} + a(\mathbf{C}^\dagger{}_x\mathbf{G}\mathbf{C}_x)\mathbf{G}\right)\mathbf{C}_y = \epsilon_2\mathbf{S}\mathbf{C}_y \qquad \langle x^4\rangle = \mathbf{C}^\dagger{}_x\mathbf{G}\mathbf{C}_x$$
$$E = \epsilon_1 + \epsilon_2 - a\langle x^4\rangle\langle y^4\rangle$$

2.3 Self-Consistent Field (SCF)

There is an additional complication with this system of eigenvalue problems. To compute $\hat{h}_{eff}(x)$, $\psi(y)$ is needed for the $ax^4\langle y^4\rangle$ part, but to get $\psi(y)$, $\hat{h}_{eff}(x)$ needs to be known, which means $\psi(x)$ is needed. To get the effective Hamiltonians, the wavefunctions of the other particles are needed, and to get the wavefunctions however, the effective Hamiltonians are needed. To address this chicken and egg problem, an iterative procedure called Self-Consistent Field procedure is outlined.

1. Start with a guess wavefunction $\psi(x)$ and $\psi(y)$. For our guess, we will use the coefficients of the ground state wavefunction from the uncoupled HO from the Variational Principle section.

2. Construct the effective Hamiltonians $\hat{h}_{eff}(x)$ and $\hat{h}_{eff}(y)$ using the guess wavefunctions/coefficients and with $\mathbf{H}, \mathbf{S}, \mathbf{G}$. Note that $\mathbf{S}, \mathbf{X}, \mathbf{H}$, and $\mathbf{G}$ are all basis dependent and doesn't need to be recomputed each iteration.

3. Diagonalize both $\hat{h}_{eff}(x)$ and $\hat{h}_{eff}(y)$ (don't forget to orthogonalize the effective Hamiltonian!). Retrieve the ground state coefficient and compute the total energy.

4. Repeat steps 2 and 3 until convergence or consistency is reached.

Convergence is reached if the difference between the previous and current iteration is below a tolerance value, or once consistency is reached. Convergence can be checked with E, $\text{norm}(\mathbf{C})$, $\text{norm}(h_{eff})$, or any other element that is changing each iteration. We will use $\text{norm}(h_{eff})$ with the frobenius norm, which is a matrix norm. (use `np.linalg.norm(A, ord='fro')`)

Here are some more explicit instructions

- Implement functions that evaluate $S_{\mu\nu}$, $H_{\mu\nu}$, and $G_{\mu\nu}$ to evaluate $\mathbf{S}, \mathbf{X}, \mathbf{H}$, and $\mathbf{G}$. Use an `n_basis >= 15` of evenly spaced gaussian functions with the same setup as before range $= [-5, 5]$, $\alpha = 2$

- Solve the uncoupled HO to get the initial guess coefficients using the code from the previous section.

- Implement the main SCF loop using the procedure in Section 2.3. Compute the final energy and wavefunction with several coupling strength parameter $a = [0, 0.1, 1, 10, 100, 1000]$ in ax^4y^4. It is also a good practice to have a maximum iteration count as a safeguard.

2.4 Correlation (Footnote)

Once again, the motion $\psi(x)$ influences $\psi(y)$ and vice versa and capturing the exact energy requires a wavefunction that exists in both coordinates $\psi(x, y)$. You can imagine that the Hartree-Product $\Psi(x, y) = \psi(x)\psi(y)$ and the mean-field approximation is inadequate because the expectation value is fundamentally independent and uncorrelated as shown in Section 2.1. This inadequacy is often referred to as correlation. To capture the energy from correlation, the wavefunction must contain some unfactorizable component $|\Psi(x, y)|^2 = |\psi(x)|^2 \, |\psi(y)|^2 + |\psi(x, y)|^2$ that allows the movement in $\psi(x)$ to correlated with the movement in $\psi(y)$.

A notable method to capturing correlation is to take linear combination of different wavefunctions or "configurations". For our coupled oscillator example, the indices ij represent higher energy wavefunctions, or columns of the coefficient eigenvector matrix corresponding to higher eigenvalues from the code above. This method is called Configuration Interactions or CI with the all possible linear combinations of configurations corresponding to Full CI.

$$\begin{aligned}\Psi(x, y) &= c_{11}\psi_1(x)\psi_1(y) + c_{21}\psi_2(x)\psi_1(y) + c_{12}\psi_1(x)\psi_2(y) + \ ... \\ &= \sum_{ij}^{N_x, N_y} c_{ij}\psi_i(x)\psi_j(y)\end{aligned}$$

In the Hartree-Fock method of molecular problems, it turns out that the mean-field approximation captures majority of the energy, often over $\sim 99\%$, but correlation still captures additional energy contributions that is often critical for the chemistry we want to simulate. CI and FCI, however, are very expensive and often an impractical way of capturing correlation do to the exponentially scaling number of determinants. (refer to Szabo chapter 2.2) Over the years, chemists, physists, and mathematicians developed, a number of successful methods that better capture correlation while maintaining practicality in terms of compute cost. Notable methods that have been developed in this pursuit include Density Functional Theory, Many Body Perturbation Theory, and Coupled Cluster Theory.

3 Additional notes

3.1 Variational Principle

Although analytically solving the Time-Independent Schrodinger Equation gives us the exact formula for the energy and wavefunctions, we often have to approximate the energy of a system with an approximate wavefunction because the analytical solution is either too difficult or impossible to evaluate. (E.g. the two electron potential $\int_{-\infty}^{\infty} dr_i dr_j 1/r_{ij}$ for electron i and j in atoms/molecules with many electrons)

The variational principle is a powerful statement/condition useful for approximating the ground state energy. The variational principle states that the trial energy eigenvalue $E[\psi_t]$ will always be greater than or equal to the true ground state energy eigenvalue E_0 of a system described by the Hamiltonian $\hat{H}$, or:

$$E[\psi] = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0$$

Visually, you can imagine that the true groundstate wavefunction is a minima of some convex energy potential well and one can achieve the best guess wavefunction eigenstate by simply minimizing the energy eigenvalue in the TISE without worrying about wandering off below the true ground state.

In practice, your Anzts (guess wavefunction) contains a single/multiple parameters that can be optimized to minimize the energy eigenvalue in the TISE. We will work exclusively with Anzts with a linear combination of basis functions:

$$|\psi\rangle = \sum_{\mu} c_\mu |\phi_i\rangle = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \begin{matrix} |\phi_1\rangle \\ |\phi_2\rangle \\ \vdots \\ |\phi_n\rangle \end{matrix} \quad \text{or} \quad \psi(x) = \sum_{\mu} c_\mu \phi_\mu(x) = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \begin{matrix} \phi_1(x) \\ \phi_2(x) \\ \vdots \\ \phi_n(x) \end{matrix}$$

where $\{\phi_1, \phi_2, ...\phi_n\}$ defines the basis set and the coefficients/vector $\{c_1, c_2, ..., c_n\}$ are the variational parameters.

Solving the TISE with this finite, coefficient vector requires transforming the Hamiltonian into a matrix via evaluating/integrating the matrix elements. This transformation can be

determined by plugging in the Anzats into the TISE:

$$\hat{H}|\psi\rangle = E|\psi\rangle \qquad |\psi\rangle = \sum_\mu c_\mu \phi_\mu(x) = \sum_\mu c_\mu |\phi_\mu\rangle$$

$$\langle\psi|\hat{H}|\psi\rangle = \langle\psi|E|\psi\rangle$$

$$\left(\sum_\mu c_\mu^* \langle\phi_\mu|\right) \hat{H} \left(\sum_\nu c_\nu |\phi_\nu\rangle\right) = E \left(\sum_\mu c_\mu^* \langle\phi_\mu|\right) \left(\sum_\nu c_\nu |\phi_\nu\rangle\right)$$

$$\sum_{\mu\nu} c_\mu^* \langle\phi_\mu|\hat{H}|\phi_\nu\rangle c_\nu = E \sum_\mu c_\mu^* \langle\phi_\mu|\phi_\nu\rangle c_\nu$$

$$\sum_{\mu\nu} c_\mu^* H_{\mu\nu} c_\nu = E \sum_{\mu\nu} c_\mu^* S_{\mu\nu} c_\nu$$

$$\mathbf{C}^\dagger \mathbf{H} \mathbf{C} = \mathbf{C}^\dagger \mathbf{S} \mathbf{C} \mathbf{E}$$

$$\mathbf{H}\mathbf{C} = \mathbf{S}\mathbf{C}\mathbf{E}$$

If we do not left multiply the resulting vector of $\mathbf{Hc}$ with $\mathbf{c}^\dagger$, we get the matrix vector equation $\mathbf{Hc} = E\mathbf{Sc}$ where $\mathbf{S} = \sum_{\mu\nu} S_{\mu\nu} = \sum_{\mu\nu} \langle\phi_\mu|\phi_\nu\rangle$ is the overlap matrix. The explicit matrix vector form of this equation looks like:

$$\begin{bmatrix} \langle\phi_1|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_1|\hat{H}|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_n|\hat{H}|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = E \begin{bmatrix} \langle\phi_1|\phi_1\rangle & \cdots & \langle\phi_1|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\phi_1\rangle & \cdots & \langle\phi_n|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$\begin{bmatrix} \langle\phi_1|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_1|\hat{H}|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_n|\hat{H}|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \langle\phi_1|\phi_1\rangle & \cdots & \langle\phi_1|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\phi_1\rangle & \cdots & \langle\phi_n|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \begin{bmatrix} E_1 & & \\ & \ddots & \\ & & E_n \end{bmatrix}$$

Note that although the overlap of the variationally minimized wavefunction $\langle\psi|\psi\rangle$ must by orthonormal, the basis functions themselves $\langle\phi_\mu|\phi_\nu\rangle$ don't necessarily have to be. This is why $S_{\mu\nu} = \langle\phi_\nu|\phi_\nu\rangle$ doesn't necessarily have to be orthonormal, but $\langle\psi|\psi\rangle = \sum_{\mu\nu} c_\mu^* \langle\phi_\mu|\phi_\nu\rangle c_\nu = \mathbf{c}^\dagger \mathbf{S} \mathbf{c} = 1$ must be true.

The general procedure for this linear variational principle method follows:

1. Select a trial wavefunction, for the Variational principle to work, the wavefunction

2. Compute (integrate) the matrix elements of $\mathbf{S}$ and $\mathbf{H}$ where $S_{\mu\nu} = \langle\phi_\mu|\phi_\nu\rangle$ and $H_{\mu\nu} = \langle\phi_\mu|\hat{H}|\phi_\nu\rangle$.

3. Minimize the eigenvalue E by solving the matrix vector equation $\mathbf{HC} = \mathbf{SCE}$ or $\mathbf{H'C'} = \mathbf{C'E}$ if orthogonalized.

3.2 Linear Variational Principle from Constrained Optimization

Another way of deriving the linear variational principle form $\mathbf{HC} = \mathbf{SCE}$ is shown here with explicitly constraining the overlap to be orthonormal $\langle\psi|\psi\rangle = \mathbf{C}^\dagger\mathbf{SC} = 1$. To minimize E with this constraint, a Lagrangian is constructed and it minimized with respect to the coefficients $\mathbf{C}^\dagger$ to find the stationary point minima:

$$\begin{aligned}
\mathcal{L} &= \langle\psi|\hat{H}|\psi\rangle - \lambda\left(\langle\psi|\psi\rangle - 1\right) \\
&= \mathbf{C}^\dagger\mathbf{HC} - \lambda\left(\mathbf{C}^\dagger\mathbf{SC} - 1\right) \\
\frac{\partial\mathcal{L}}{\partial\mathbf{C}^\dagger} &= 0 \qquad \frac{\partial\mathcal{L}}{\partial\lambda} = 0 \\
\frac{\partial\mathcal{L}}{\partial\mathbf{C}^\dagger} &= \mathbf{HC} - \lambda\mathbf{SC} = 0 \\
\mathbf{HC} &= \lambda\mathbf{SC}
\end{aligned}$$

If the multiplier $\lambda = E$, then the generalized linear variational principle equation $\mathbf{HC} = \mathbf{ESC}$ is obtained.

3.3 Orthogonalization

To solve the $\mathbf{HC} = \mathbf{SCE}$ as an eigenvalue problem, one can modify the equation to $\mathbf{S}^{-1}\mathbf{HC} = \mathbf{CE}$. However, this requires a direct inversion of $\mathbf{S}^{-1}$ which increases/amplifies the numerical errors if $S_{\mu\nu}$ is small (aka the basis functions are far apart with little overlap). This direct inversion can be avoided by constructing some orthogonalizing transformation $\mathbf{X}$ such that $\mathbf{X^T SX} = \mathbf{I}$. (You can see that the overlap matrix of an orthonormal basis is the identity matrix) With such transformation, the overlpan matrix disappears and the equation is transformed into $\mathbf{H'C'} = \mathbf{C'E}$ where $\mathbf{H'} = \mathbf{X^T HX}$ and $\mathbf{C} = \mathbf{XC'}$.

One possible choice of such transformation, and a surprisingly simple choice, is $\mathbf{X} = \mathbf{S}^{-1/2}$. Plugging this into the equation above that this does indeed orthogonalize the basis vectors: $\mathbf{X^T SX} = (\mathbf{S}^{-1/2})^\mathbf{T}\mathbf{SS}^{-1/2} = \mathbf{S}^{-1/2}\mathbf{SS}^{-1/2} = \mathbf{S}^0 = \mathbf{I}$. Once again, to avoid the direct inversion of $\mathbf{S}$, we will transform it into the diagonal matrix $\mathbf{s}$ for the inversion and transform it back into $\mathbf{X}$.

Here is a more explicit proceedure for building and using an orthogonaling transofrmation:

1. Diagonalize $\mathbf{S}$ to find diagonalizing transformation (eigenvectors) $\mathbf{U}$ and the diagonals $\mathbf{s}$ (eigenvalues). ($\mathbf{SU} = \mathbf{Us}$)

2. Take the inverse square root of the diagonals $\mathbf{s}^{-1/2}$ and transform back into the original basis to find $\mathbf{X}$. Since $\mathbf{S} = \mathbf{UsU^T}$ (right multiply $\mathbf{U^T}$ to the eigenvalue equation $\mathbf{SU} = \mathbf{Us}$), $\mathbf{X} = \mathbf{Us}^{-1/2}\mathbf{U^T}$.

3. Transofrm the Hamiltonian into the orthogonal basis $\mathbf{H}' = \mathbf{X^T HX}$ and diagonalize the orthogonalized Hamiltonian to obtain the energies $\mathbf{E}$ and the orthogonalized coefficients $\mathbf{c}'$. ($\mathbf{H}'\mathbf{C}' = \mathbf{C}'\mathbf{E}$)

4. Transform the coefficient back into the original basis $\mathbf{C} = \mathbf{XC}'$ to obtain the coefficients $\mathbf{C}$ back.